ContentCraft: Web App for Social Media Content Generation

DEPARTMENT OF INFORMATION TECHNOLOGY
UNIVERSITY OF SARGODHA
SARGODHA – PAKISTAN

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ContentCraft: Web App for Social Media Content Generation

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FINAL APPROVAL

The final approval page will be provided by the department after the final evaluation.

DEDICATION

With warm hearts and deep respect, we humbly dedicate this Final Year Project to our beloved parents. Their endless love, constant support, and sincere prayers have been the strength behind all our efforts. They have stood by us in every step, cheering us on with patience and hope. Without them, this journey would not have been possible. As the saying goes:
"Parents are the anchors of a child's life and the wind beneath their wings."

We are also truly thankful to our respected teachers, who have guided us with care and dedication. Their encouragement and knowledge helped shape our ideas and efforts throughout these years.
"A good teacher is like a candle – it consumes itself to light the way for others."
To all our teachers and mentors, thank you for believing in us and helping us grow.

Lastly, this project also stands as a testament to our teamwork, mutual respect, and perseverance. We are proud of what we have accomplished together. This journey has not only been academic but a memorable chapter of learning and growth. This project is more than just a requirement—it's a special memory and a symbol of how far we've come.
"Coming together is a beginning; keeping together is progress; working together is success." – Henry Ford

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We are thankful to all our teachers and mentors at the **Department of Information Technology, University of Sargodha**, for equipping us with the knowledge and tools that shaped our academic growth.

Finally, we express our deepest appreciation to our families for their constant love, support, and patience, and to each other as teammates — for the cooperation, understanding, and commitment that brought ***ContentCraft*** to life.

PROJECT BRIEF

PROJECT NAME	CONTENTCRAFT: WEB APP FOR SOCIAL MEDIA CONTENT GENERATION
ORGANIZATION NAME	UNIVERSITY OF SARGODHA
OBJECTIVE	TO DESIGN AND DEVELOP A WEB-BASED APPLICATION THAT ASSISTS USERS IN GENERATING HIGH-QUALITY SOCIAL MEDIA CONTENT USING AI-POWERED TOOLS
UNDERTAKEN BY	MUHAMMAD ABRAR AHMED TUBA ARIF RABIYYA ARSHAD
SUPERVISED BY	SIR AZHAR MUSHTAQ ASSISTANT PROFESSOR DEPARTMENT OF INFORMATION TECHNOLOGY UNIVERSITY OF SARGODHA
STARTED ON	SEPTEMBER 15, 2024
COMPLETED ON	MAY 26, 2025
COMPUTER USED	HP ELITEBOOK 840 G3
SOURCE LANGUAGE	REACT.JS & NODE.JS
OPERATING SYSTEM	WINDOWS 10
TOOLS USED	VISUAL STUDIO CODE, MYSQL DATABASE

ABSTRACT

In today's fast-paced digital era, video has become the dominant form of content across platforms, yet much of its potential remains untapped when it comes to written communication. **ContentCraft** is a web-based application developed to bridge this gap by converting video audio into effective, text-based content tailored for social media platforms.

Aimed at influencers, educators, journalists, news agencies, and blog writers, ContentCraft offers a seamless solution for repurposing spoken words into engaging captions, tweets, Facebook posts, and other text formats. The platform automatically transcribes video audio and generates well-structured, platform-appropriate content—eliminating the need for manual transcription and copywriting.

By enabling users to transform their video content into shareable written material quickly and accurately, ContentCraft enhances productivity, ensures consistency across platforms, and supports the growing demand for rapid, high-quality content distribution. It is a timely tool for creators seeking to extend the impact of their voice across the digital ecosystem with minimal effort and maximum reach.

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1 Introduction

1.1 System Introduction

ContentCraft is a one-of-a-kind AI assistive tool designed for social media content creation. It specifically targets motivational authors, influencers, bloggers, journalists, and educators, enabling them to generate creative content ideas efficiently from the audio of their videos. The platform is web-based and processes uploaded video files by extracting the audio, transcribing it into text, and using that text to suggest web content ideas. By integrating transcription with creative content support, ContentCraft reduces the manual workload of content creators and allows them to focus more on their content strategy. It aims to enhance productivity, streamline idea generation, and encourage smart content planning by leveraging artificial intelligence.

1.2 Background of the System

Existing tools like Otter.ai handle transcription, while BuzzSumo and SocialBee assist with content ideation and scheduling. However, these tools work in isolation, requiring users to switch between platforms. ContentCraft bridges this gap by offering a unified web-based solution that transcribes video audio and generates relevant content ideas in one place. This integration streamlines the workflow, boosts productivity, and simplifies the content creation process.

1.3 Objectives of the System

- To create convenience for the users in terms of uploading videos and extracting audio from the videos.
- To correctly convert the audio of the video into text for analysis using a powerful model.
- To improve the efficiency of transcribing an entire audio, by applying audio processing to noise reduction and time gaps removal.
- To Provide a recommendation for content that will please the user's audience and meet the goals.
- To manage user profiles and ensure data security across the platform.

1.4 Significance of the System

ContentCraft plays a very important role in smoothing the content creation process. By automating transcription and content ideation. The tool is significant in multiple application areas, including:

- **Motivational Influencers:** Helping to come up with compelling captions and video titles or blog posts from speeches or videos aired.
- **Educators:** Assisting in the generation of lesson plans, articles for blogs, study guides generated from lecture videos.
- **Bloggers & Journalists:** Providing timely tips on the articles or reports ideas based on the interview or event recorded.

2 Overall Description

2.1 Product Perspective

ContentCraft is an AI-powered web application designed to simplify social media content creation by combining audio transcription and content idea generation in one platform. Unlike existing tools that require multiple applications, ContentCraft streamlines the process by using integrated AI models to convert video's audio into text and generate relevant content suggestions. All user data is securely stored in a database, offering a seamless and efficient experience for content creators.

2.2 Product Scope

ContentCraft focuses on automating the conversion of video's audio into text and generating creative content ideas for use in social media, blogging, and educational contexts. The tool streamlines the content creation process by offering features like video upload, audio transcription with noise reduction, and AI-driven idea generation—all within a single platform. Its core scope includes ensuring transcription accuracy, delivering relevant content suggestions, and managing user access and data through an integrated user administration system.

In Scope:

- Video upload and audio extraction from the video.
- Transcription of uploaded video's audio into text.
- Content idea generation using transcribed text through video.

Out of Scope:

- Transcription of live streaming video and audio.
- Advanced video editing.
- Content posting (e.g., directly to social media).

2.3 Product Functionality

- **Video Upload & Audio Processing:** Allows users to upload videos and automatically process the audio. The system detects distortions or silent segments to improve audio quality.
- **Transcription:** Converts processed audio into text with enhanced accuracy by applying noise reduction techniques.
- **Content Idea Generation:** Analyzes the transcribed text using AI models to generate relevant and platform-specific content suggestions.

2.4 Users and Characteristics

There are multiple user types for ContentCraft, with varying characteristics:

Motivational Influencers:

- Usage Frequency: High as content is created relatively frequent on this site.
- Technical Expertise: Average as they familiar with basic tools but not advanced tech skills.
- Functions Used: Video sharing, captions and the use of ideas for content.
- Importance: High as the content of media highly depends on the idea creation process which is to happen on a continuous basis.

Bloggers/Journalists:

- Usage Frequency: Moderate to High (Depending on mean and frequency of interviews, videos post).
- Technical Expertise: Moderate – generally comfortable with transcription tools and basic editing platforms.
- Functions Used: Transcription, content suggestion, managing the users.
- Importance: High as it depends on the transcriptions and on content for the idea.

Educators:

- Usage Frequency: Low to medium (For content development and video transcription).
- Technical Expertise: Low (familiar with simple interfaces).
- Functions Used: It is about transcription and content generation.
- Importance: Moderate as they demand correct transcriptions and ideas about the data in a video.

Most Important Users: Motivational Influencers, Bloggers, and Journalists are the target users because they depend mostly on transcription proficiency and content creation.

2.5 Operating Environment

ContentCraft is designed to operate across multiple computing platforms with a focus on web compatibility and modern browser support. It is a web-based application accessible through standard desktop and mobile browsers and hosted on a backend server environment.

Minimum Hardware Requirements:

- Processor: Intel Core i5 or equivalent.
- Memory: Minimum 8GB RAM.
- Storage: Minimum 256GB SSD.

Software Requirements:

- Operating System: Windows, macOS, or Linux.
- Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge (latest versions).
- Backend: NodeJS environment.
- Frontend: ReactJS-based user interface.
- Models: For Transcription and Content Idea Generation.
- Other Tools: FFmpeg for audio extraction and processing.

The software has to coexist smoothly with Model services and take the weight of data processing (transcription, content creation) and should work well within todays web compliance standards.

3 Specific Requirements

3.1 Functional Requirements

In this section, we'll break down the specific functions of ContentCraft and organize them by functional areas:

3.1.1 Video Upload & Audio Processing

Video Upload:

- Uploaded video files must be easily understandable by users and should be allowed from common formats such as MP4.
- It should also check the file size and format before it process it.

Audio Extraction:

- The FFmpeg package for extraction of audio part from the uploaded video will be used in the current system.
- The system is possible to track and dispel the noise, the interference or the empty space of the audio.
- If after processing the audio contains some quality issues the user will be notified.

3.1.2 Transcription & Text Conversion

Audio Transcription:

- The extracted audio will be converted into text using the model.

- The transcription process must deal with different languages supported by the model.
- If transcription fails, the user must be notified with error messages.

Text Conversion:

- Transcribed text will be saved for more processing and can be copied by the user.
- The text can be change within the platform for user corrections as per requirements.

3.1.3 Content Idea Generation

AI-Driven Content Suggestions:

- The transcribed text will be processed by the model to generate content ideas.
- Recommendations will be based on a user's provided objective (e.g. motivational messages, educational messages).
- Users can provide input parameters, such as numbers of posts, hashtags & platforms.

3.1.4 User Management

User Registration & Authentication:

- Users can register, log in, and manage their profiles without any privacy concern.
- The system will provide access control based on user authentication

Data Management:

- All transcriptions, video uploads, and content suggestions will be stored securely in a database.
- Users can view and manage their previous uploads and generated content through a dashboard.

3.1.5 Data Storage

Data Retrieval:

- Users can retrieve their stored transcriptions and content ideas at any time.

3.2 Behaviour Requirements

Actors:

- User: (Influencer, Blogger, Educator) Accountable for the log in and sign up, uploading videos, asking to process audio, request to system to create ideas on contents.
- System: Takes care of converting the audio that can be taken out from the video uploads into text format. The transcriptions are analyzed to come up with content that will be made in response to the user request.

Use Case Summary:

- Login & Signup: The user simply registers or login in our platform.
- Upload Video: The user uploads a video, which is processed by the system to extract audio.
- Process Audio: For instance, the system employs FFMPEG to extract sound from the uploaded video clip.

- Transcribe Audio: The extracted audio is then taken through model and transcribed.
- Generate Content Ideas: The transcription is then fed to another model where the program get content creation ideas from the transcribed text.
- Store Data: Ideas for content generation and the content transcriptions are then saved in the database for future use.

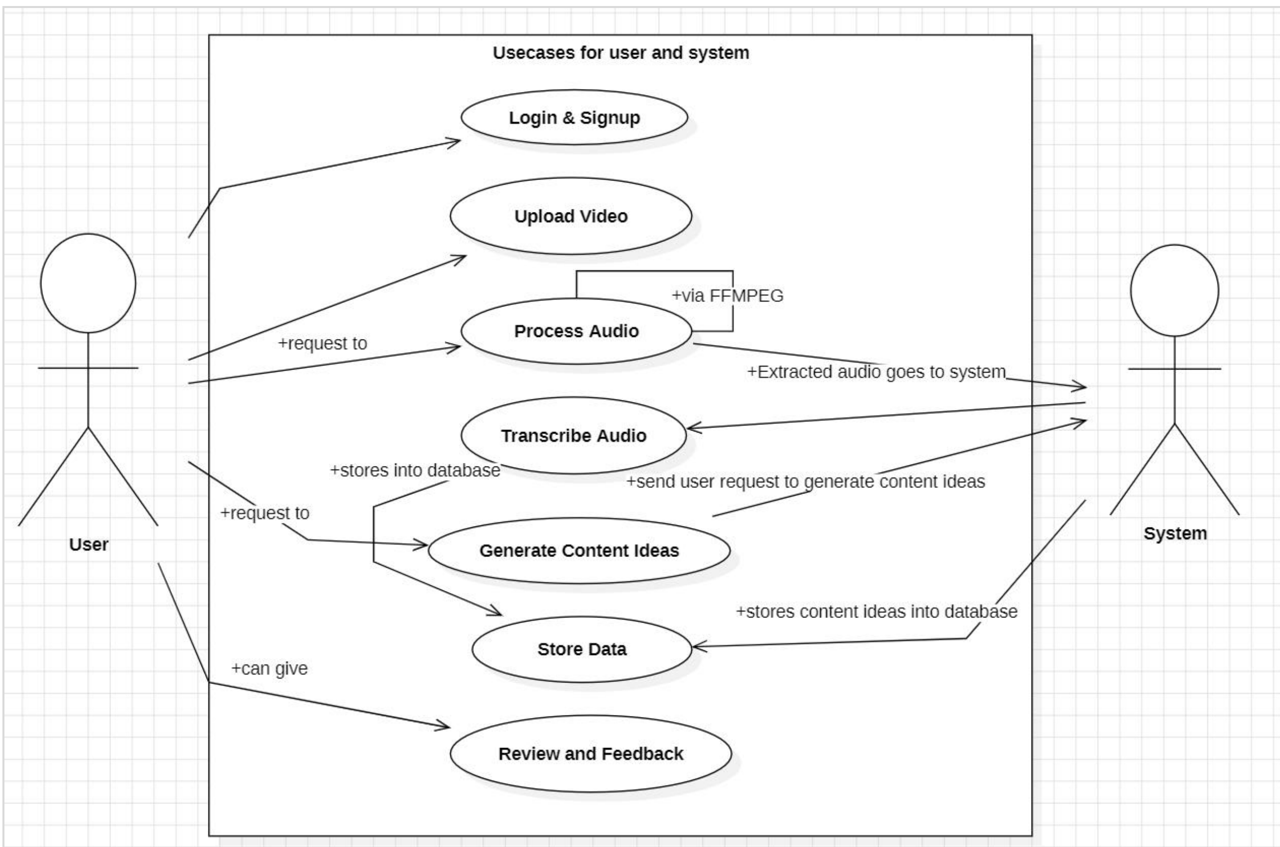

Fig 3.1 UseCase Diagram

3.3 External Interface Requirements

3.3.1 User Interfaces

ContentCraft will have a clean and easy to understand interface that will be supported on web browsers. Below are the key UI components:

Login/Registration Screen:

- Fields: Username, Password, Email, Sign-up/Login buttons.
- Error messages for incorrect credentials.

Dashboard:

- Displays user profile details and links to key functionalities: upload video, view previous transcriptions, and view already generated content ideas.
- Buttons: “Upload Video,” “View Transcriptions,” “View Content Ideas.”

Video Upload Screen:

- Fields: Video file upload, progress bar, and error handling (e.g., “Unsupported file format”).
- Buttons: “Upload,”.

Transcription Result Screen:

- Displays the transcription text with copied.
- Buttons: “Edit,” “Copied,”

Content Idea Generation Screen:

- Displays generated content ideas.
- Fields: Input platform, number of posts & hashtags.
- Buttons: “Generate Ideas”, “Copied”.

3.3.2 Hardware Interfaces

ContentCraft needs to interface with the user system for uploading of video files and handling of audio which occurs on the server. Here’s the hardware interface:

- Processor: The web application server will use Intel i5 (or equivalent) to handle multiple requests.
- Memory: 8GB RAM to ensure smooth operation of same processes (video upload, transcription, and AI generation).
- Storage: The system must handle video file storage efficiently using 256GB SSD.

FFmpeg will be used to interface with the audio processing system.

3.3.3 Software Interfaces

The software components for ContentCraft include:

- Model 1: Used for transcribing audio. The system communicates with the model by sending audio data and receiving the transcription in return.
- Model 2: The transcribed text is passed to the model to generate content ideas.

- Database: The database stores user profiles, video files, transcriptions, and content ideas. The backend uses NodeJS to interact with the database through queries.
- Operating System Compatibility: The system can work with Windows, macOS, and Linux.

3.3.4 Communications Interfaces

- HTTPS for secure web communication
- RESTful MODEL for client-server communication

4 Other Non-functional Requirements

4.1 Performance Requirements

ContentCraft's performance requirements are as follows:

- Uploading Video: Audio from a video should processed and taken in less than a minute.
- Transcribing Audio: In 45 seconds, a 5-minute audible clip could transcribed.
- Content Creation: The model should give content ideas in 30 seconds.
- System Reaction Time: Response times for frontend activities as clicks should be done in less than a minute.
- Database Management: Queries must be finished in 10 seconds.

Example Scenario: A user uploads a video, which is processed in 30 seconds, transcribed in 45 seconds, and content suggestions are given to them in 30 seconds. All of this got done while the system operates accurately and the database is accessed quickly.

4.2 Safety and Security Requirements

Privacy of Data is one of the issues that are important to our organization. Here are the necessities filed below:

- Data Security and Storage: In order to determine approach, all the data of the user including the videos of the user, transcriptions and content ideas should be stored in a safe zone, transcriptions, and content ideas—must be preserved in a secure region.

- User Verification: To be able to access the site, the user is required to list a secure login and identification.
- Encryption: The success SSL/TLS codes to encrypt all the data being shipped from two points the frontend and the backend.
- Backup of Data: For the purpose of avoiding losing of data, systematic data backup should be done as suggested.
- Control of Access: Table orientations as well as sensing disturbances is merely do by distinct intend directors.
- Observance of the Rules: Hence to ensure the privacy of the user, the system should take a responsibility (to follow the GDP).

4.3 Software Quality Attributes

4.3.1 Reliability: The system should be super reliable and maintain a highly uptime. This means it should be operational most of the times.

4.3.2 Portability: They have to ensure that the software performs well on the operating systems that are widely used commonly included in Windows OS, Mac OS and Linux. We shall ensure that we implement tools and frameworks that are compatible across the systems.

4.3.3 Maintainability: There is no way that in future you won't want to change something in the code and thus it has to be easy to maintain. That's why it will be clean, modular code that will write together with comprehensive comments for other people to read.

4.3.4 Usability: The used system should not be complicated and should be easily understandable even if the user had no previous experience with the system.

4.3.5 Adaptability: In this case, the software has to always be responsive to trends and users' necessities. We want to make it fully expandable so that adding new features or updating the existing system won't be difficult.

4.3.6 Flexibility: It should do multiple purposes, the use cases where it can generate different contents like text for motivational or educating videos.

4.3.7 Interoperability: We also hope that the software will be able to work with other MODELS.

4.3.8 Testability: It is important that each of the parts that constitute the system can be easily tested.

4.3.9 Reusability: Our goal will be to make the code reusable for other projects. By keenly separating the components and making them as independent.

4.3.10 Robustness: Special attention should be paid for its ability to deal with such conditions of disturbances well. It will also have good error handling and validation checks so that we can be sure it can handle errors or take in bad input.

4.3.11 Availability: Ideally, the software should be running all the time and, therefore, easily accessible by the users. Here, some of the methods to achieve this include load balancing where the systems function as expected despite lots of people using it at one time.

5 Design Description

5.1 Composite Viewpoint

Common Utilities include tools used by one or many apps in the system, Video Upload & Audio Processing include the process of uploading videos and extracting audio from them. It transcribes audio to produce text thus enabling many individuals with special needs to access the content. Content Idea Generation provides topics and keyword extracted from the text for content generation purposes. User Management & Database authority is correspondent to user authentication and authorization while the database is responsible for storing data. In each of the modules, integration with Common Utilities with respect to a good flow of data and process connectivity.

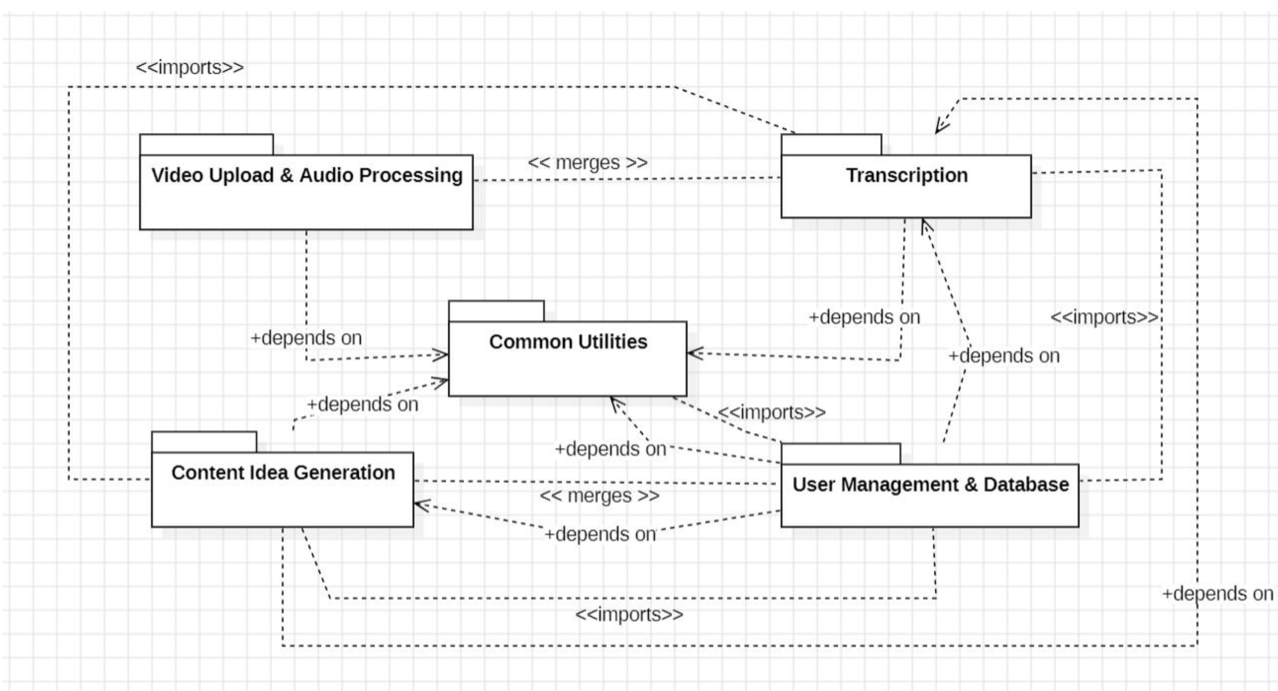

Fig: 5.1 Package Diagram

5.2 Logical Viewpoint

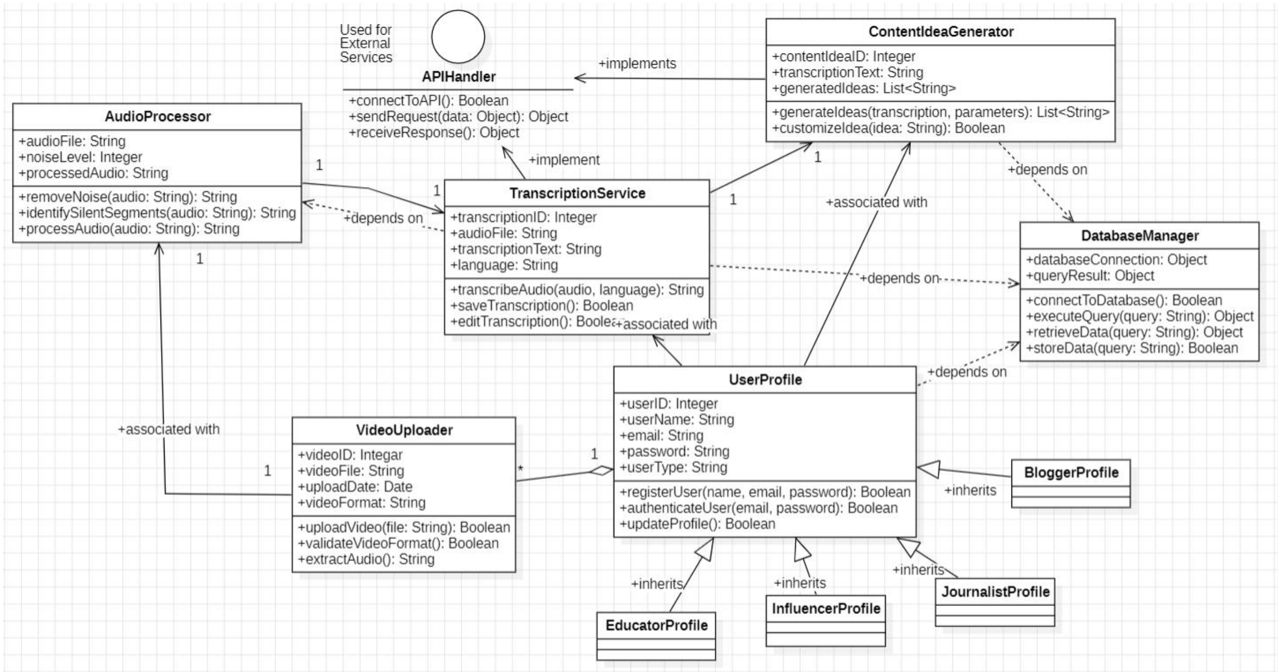

Fig 5.2 Class Diagram

5.3 Information Viewpoint

This is described in the following ERD of ContentCraft.The User table includes a userID and the user's email address linking to Video, which stores videos that the user uploads. Every Video is related to Audio, which is responsible for the extracted audio information. The Transcription table has transcribed text that is related to the audio through foreign keys. Last of it, ContentIdea comes up with ideas from transcriptions and refers learners to transcriptions. Thus, there is the relational structure of operations providing the means for the efficient information exchange from user uploads to content ideas generation.

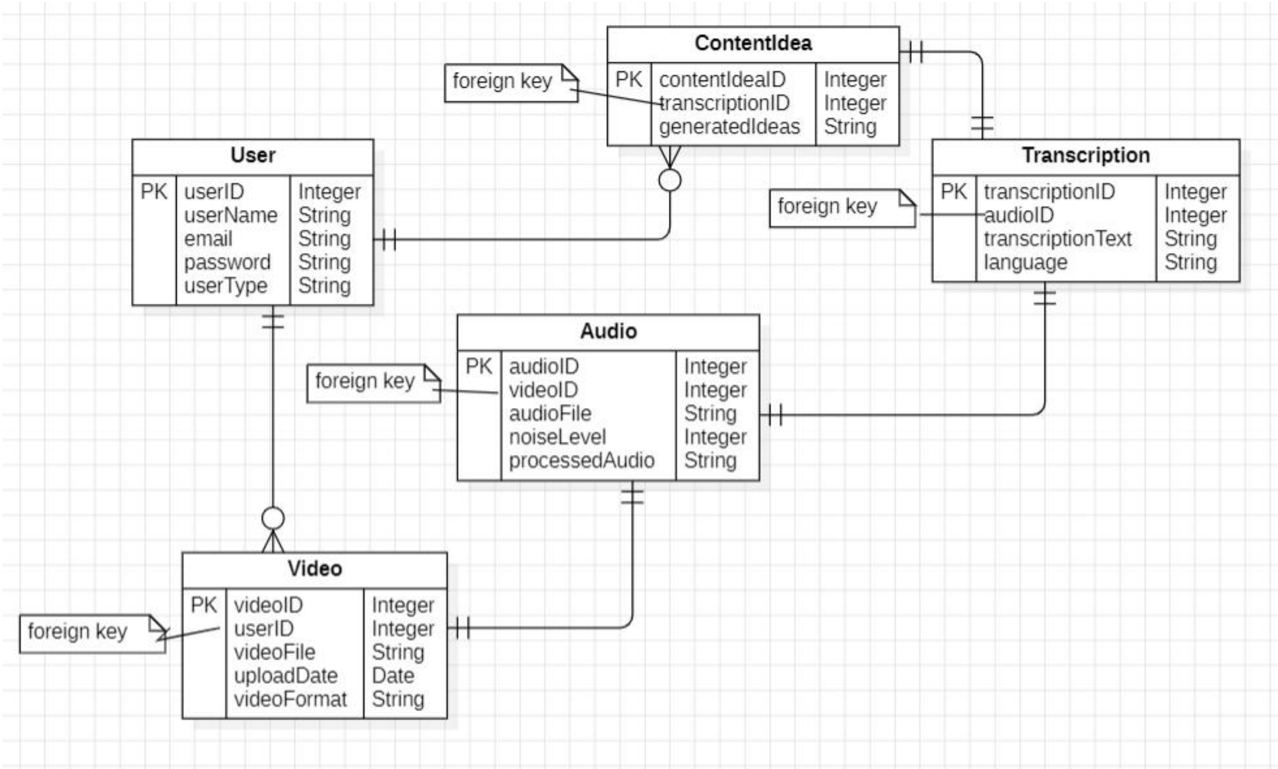

Fig 5.3 ERD

5.4 Interaction Viewpoint

The sequence diagram of the ContentCraft integrates the end-to-end process of handling the video content. User uploads a video in the system and the VideoUploader of the system forwards it to the transcriber so that it can extract audio from the video. After that, the Transcribed text utilizes the Model to turn the recorded audio to text. The transcribed text, as you will remember, is saved by the DB Manager (SQL). Last of all, the ContentIdeaGenerator utilizes this text to give content ideas and that makes up the Model. This makes the handling process swift right from the moment that a video is uploaded to a time a creative idea is conceived.

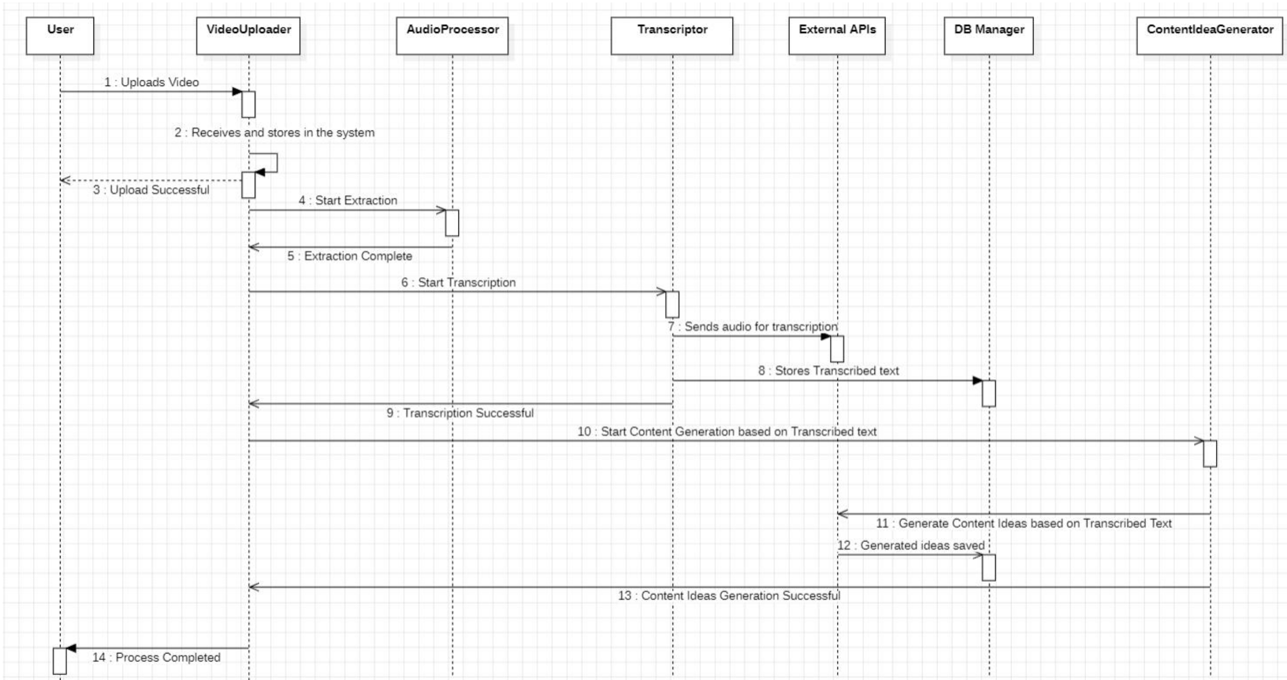

Fig 5.4 Sequence Diagram

5.5 State Dynamics Viewpoint

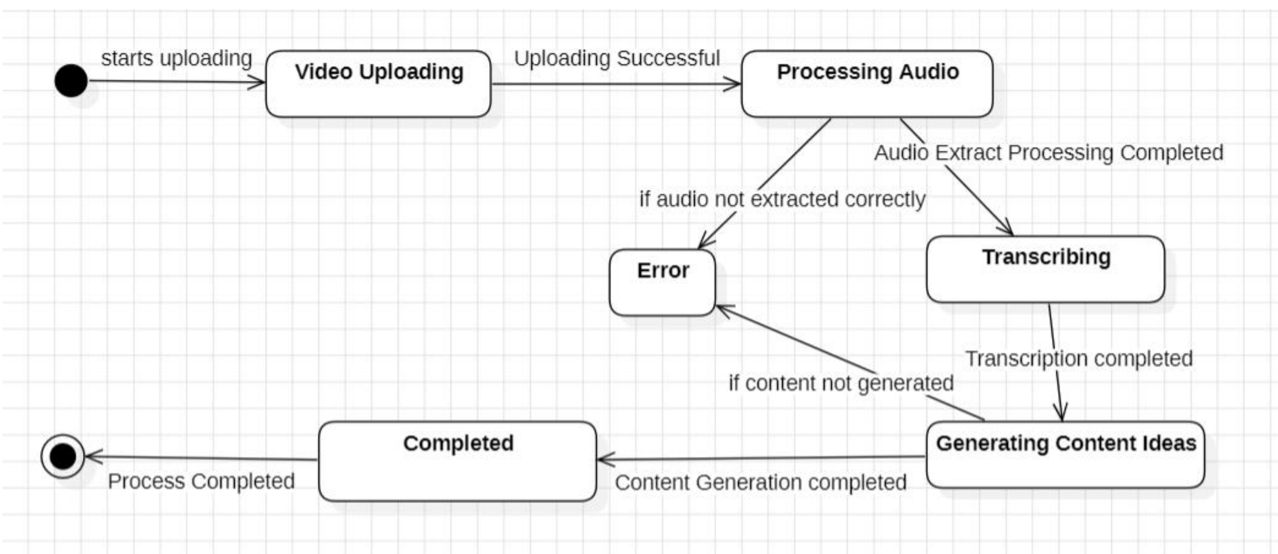

Fig: 5.5 State Machine Diagram

Current State	Event	Next State	Action
Idle	User uploads video	Uploading	Start video upload
Uploading	Video upload completed	Processing Audio	Start audio extraction
Processing Audio	Audio processing completed	Transcribing	Convert audio to text
Transcribing	Transcription completed	Generating Content Ideas	Start content idea generation
Generating Content Ideas	Idea generation completed	Completed	Display results
Any	Error occurs	Error	Display error message

Table: 5.1 State Matrix Table

5.6 Algorithm Viewpoint

```
Function TranscribeAudio(videoFile):
    # Step 1: Extract audio from video
    audioFile = ExtractAudio(videoFile)

    # Step 2: Transcode audio into required format
    transcodedAudio = TranscodeAudio(audioFile)

    # Step 3: Detect the language of the audio
```

```
    language = DetectLanguage(transcodedAudio)

    # Step 4: Transcribe the audio into text using Google
MODEL
    transcription = GoogleSpeechToTextMODEL(transcodedAudio,
language)

    # Step 5: If transcription is successful
    If transcription.success:
        Return transcription.text
    Else:
        RaiseError("Transcription Failed")
EndFunction

Function GenerateSocialMediaPost(transcriptionText,
targetLanguage):
    # Step 1: Translate the transcription into the target
language
    translatedText = TranslateText(transcriptionText,
targetLanguage)

    # Step 2: Generate a social media post based on the
translated text
    postContent = OpenAIGeneratePost(translatedText)

    # Step 3: Return the generated post
```

```
    Return postContent
EndFunction
```